

AI Data Augmentor - Assignment #3 Reflection

Aleks Obuhovskiy | Automated company-data enrichment with human quality assurance

Objective

The goal was to enrich a starter list of 50 apparel and outdoor companies with a location, customer-service phone number, and official website. I built a Python workflow to automate the research while avoiding guessed or hallucinated values.

Tool Stack

Python	automation and validation logic
DDGS	free web-search discovery
requests + BeautifulSoup	direct official-site retrieval
pandas + openpyxl	CSV/Excel processing
phonenumbers	phone parsing and normalization
VS Code + GitHub	development and version control

Execution Strategy

- Read the starter CSV and process each company.
- Discover likely official domains and reject retailers, directories, social sites, regional duplicates, and suspicious look-alike domains.
- Search official-site content for customer-service phone and headquarters/corporate-location evidence.
- Use caching, retries, phone-context scoring, and an UNKNOWN fallback when evidence is not strong enough.
- Write the enriched data to Excel and preserve source references for QA.

Challenges and Quality Control

The main challenge was not finding data, but deciding which result was trustworthy. During testing, search results sometimes returned a regional site instead of the primary site, a retailer instead of the brand, a warranty number instead of customer service, or a distribution/office location instead of headquarters. Examples included Arc'teryx regional domains, Gregory retailer results, and Mountain Hardwear warranty-phone confusion.

I improved the workflow with stronger domain validation, phone-context ranking, location rules, caching, and repeated QA tests. I also performed a final human review against official company sources. When the automation produced a questionable result, I corrected it only when stronger official evidence was available. If a value could not be verified reliably, I kept **UNKNOWN** rather than guessing.

Outcome / Lesson Learned

The project showed that AI-assisted data augmentation can save significant research time, but web data still requires governance and human review. The most important lesson was that accuracy depends on validation rules and source quality, not simply on returning a value for every field. The final deliverable therefore combines automated enrichment with transparent human QA.